MACHINE LEARNING ASSIGNMENT

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A1. What is Machine Learning?

Machine learning is a type of artificial intelligence (AI) that allows computers to learn from data and identify patterns so that they can make predictions or decisions without a person having to program every rule manually.

For example, a school can use machine learning to predict whether a learner is likely to pass or fail an examination. The computer can learn from previous learners' information, such as the number of hours they studied, their attendance, and whether they passed or failed previous class tests. After learning from this data, the computer can use the patterns it has discovered to predict the results of new learners.

- Difference between Traditional Programming and Machine Learning

Traditional programming is when a person writes specific instructions or rules for a computer to follow. The computer applies those rules to the data to produce an answer.

Machine learning is different because the computer is given data and, in supervised learning, examples of the correct answers. It uses these examples to identify patterns and learn rules that can be used to make predictions about new data.

- Traditional Programming

Data + Rules written by a person



Computer processes the instructions



Answer

- Machine Learning



Training data + Correct answers



Computer learns patterns and builds a model



Model predicts answers for new data

A2. How does machine learning relate to artificial intelligence, deep learning and data science?

2.1 Artificial Intelligence (AI)

Artificial intelligence is the field of technology that enables computers to perform tasks that normally require human intelligence, such as understanding language, recognising images and solving problems. e.g ChatGPT

2.2 Machine Learning (ML)

Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data and use those patterns to make predictions or decisions without needing every rule to be programmed manually.E.g A computer learning from previous learners' results to predict whether a new learner will pass.

2.3 Deep Learning (DL)

Deep learning is a specialised type of machine learning that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data. E.g A computer recognising faces in photographs.

2.4 Data Science

Data science is the field of collecting, cleaning, analysing and interpreting data to find useful information and support decision-making. It can use statistics, visualisation and machine learning. E.g A school analysing learners' marks and attendance to identify why some learners perform poorly

A3. Define the core terms each with an example: dataset feature, label(target)model training predictions.

3.1 Feature

A feature is an input variable that provides information the computer uses to learn patterns and make predictions.

Example: In the dataset, Study hours and Attendance (%) are features because they may help the computer predict whether a learner will pass.

3.2 Label (Target)

A label, also called a target, is the correct answer that a supervised machine learning model learns to predict.

Example: Passed is the label because it contains the answers Yes or No. The computer uses these known answers to learn from previous learners.

3.3 Model

A model is the result of the training process. It learns patterns from the dataset and uses them to make predictions about new information.

Example: After learning from previous learners' study hours, attendance and results, the model may learn patterns associated with passing or failing.

3.4 Training

Training is the process of teaching a machine learning algorithm to identify patterns in data.

Example: The computer studies the previous learners' study hours and attendance alongside their known examination results. It uses this information to learn how these factors relate to passing or failing.

3.5 Prediction

A prediction is the answer produced by a trained model when it receives new information.

Example: A new learner studies for 7 hours and has 90% attendance. The model might predict `Yes`, meaning the learner is likely to pass. However, this prediction is not a guarantee of the actual result.

A4. Find three real-world uses of machine learning in South Africa or elsewhere in Africa. For each: what data goes in, what the model predicts, and who benefits. Ideas: fraud detection in mobile money, credit scoring, crop or livestock disease detection, product recommendations, predicting network faults

Application	Data that goes in	What the model predicts	Who benefits
Mobile Money Fraud detection (West Africa0	Transactions and scam-related patterns	Suspicious or potentially fraudulent activity	Customers and financial institutions
Credit scoring (South Africa)	mobile usage patterns, airtime usage, call data records and other available credit information.	Credit risk and likelihood of repayment	anks, lenders and people with limited credit histories who may otherwise struggle to access loans.
Crop disease detection	Photographs of crop leaves taken using a smartphone.	Whether the plant shows signs of disease or pests and, where supported, which disease may be present.	Farmers, agricultural extension workers and farming communities, because early detection can help them take

			action before crop losses increase.

Conclusion: These examples show how machine learning can solve practical problems in Africa. It helps financial institutions detect fraud, helps lenders assess credit risk, and helps farmers identify crop diseases. In each case, the model learns patterns from data and uses them to make predictions that support better decisions.

A5. Describe the typical machine learning workflow from start to finish Problem definition → collecting data → preparing data → choosing and training a model → evaluating it → using (deploying) it. Explain why the data preparation step usually takes the most time.

51 Problem definition

This is the first step, where we identify the problem, we want machine learning to solve and decide what the model should predict.

Example: A school wants to predict whether a learner will pass or fail based on study hours and attendance.

5.2 Collecting data

In this step, we gather the information needed to solve the problem. Data can come from school records, surveys, databases or other sources.

Example: The school collects previous learners' study hours, attendance percentages and examination results.

5.3 Preparing data

Data preparation involves cleaning and organising the collected data so it can be used to train a machine learning model.

This step may include:

- Removing duplicate records.
- Handling missing values.
- Correcting incorrect information.
- Converting text values into numerical values when required by the algorithm.
- Selecting relevant features and identifying the target.
- Splitting the data into training and testing sets

5.4 Choosing and training a model

In this step, we choose a suitable machine learning algorithm and train it using the prepared training data.

Example: The school chooses a classification algorithm because it wants to predict one of two categories: `Yes` or `No`.

The algorithm learns patterns from study hours, attendance and the known examination results. After training, it produces a model that can make predictions.

5.5 Evaluating the model

Evaluation checks how well the trained model performs on data that was not used to train it.

We compare the model's predictions with the actual results to see whether it makes reliable predictions.

Example: The school tests the model using learner records that were kept separate during training.

For a classification model, the school could measure accuracy, precision, recall and F1-score. This helps determine whether the model is suitable for its intended purpose.

5.6 Using or deploying the model.

Deployment means making the trained model available for use in a real-world situation.

Example: The school uses the model to predict whether a new learner is likely to pass based on their study hours and attendance.

The school can use the prediction to identify learners who may need extra support. The prediction should not be treated as a guaranteed result.

The model should also be monitored to check whether its predictions remain reliable as new data becomes available.

Part B of 4 Supervised VS Unsupervised

B1 What is Supervised Learning?

Supervised learning is a type of machine learning in which a computer learns from labelled data to make predictions about new information. The computer is given examples containing both the input information and the correct answers so that it can learn patterns from them.

- What does “supervised” mean?

The word supervised means that the computer learns from examples where the correct answers are already known. These answers guide the learning process, helping the model learn the relationship between the input data and the expected output.

For example, a computer can learn from previous learners' study hours, attendance and examination results to predict whether a new learner will pass or fail.

- What is labelled data?

  Labelled data is data that contains input information together with the correct answer, also called the label or target

B2. What is unsupervised learning? What the algorithm does when there are no labels tolearn from, and one everyday example.

Unsupervised learning is a type of machine learning in which a computer learns from data that has no labelled answers. Instead of being told what the correct answer is, the algorithm looks for patterns, similarities and groups within the data.

- What happens when there are no labels?

When there are no labels, the computer does not have a correct answer to learn from. Instead, it analyses the available features to discover patterns or organise similar records into groups.

For example, the algorithm might group customers who frequently buy similar products together, without being told beforehand which customers belong in each group.

B3. Compare the two in a table. Rows: goal · type of data needed · typical tasks ·example algorithms · how success is measured · one real-world example of each

Comparison	Supervised Learning	Unsupervised Learning
Goal	To learn from examples with known answers and predict the correct answer for new data.	To discover patterns, similarities, or groups in data without known answers.

Type of data needed	Labelled data, meaning the data includes features and a known target or answer.	Unlabelled data, meaning the data has features but no known target or answer
Typical task	Classification (predicting categories) and regression (predicting numerical values	Clustering (grouping similar items) and anomaly detection (finding unusual patterns).
Example algorithms	Decision Tree, Logistic Regression, Linear Regression, and Random Forest.	K-Means Clustering, Hierarchical Clustering, and DBSCAN.
How success is measured	By comparing predictions with known answers using measures such as accuracy, precision, recall, F1-score, or mean squared error	By checking how well the data is grouped or how useful the discovered patterns are, using measures such as silhouette score or expert judgement.
One real-world example	Learner pass prediction: A school uses learners' study hours, attendance, and previous results, together with known pass/fail outcomes, to predict whether a new learner is likely to pass.	Supermarket customer grouping: A supermarket groups customers based on their shopping habits to identify similar customers and provide relevant promotions.

B4. Explain classification and regression (both supervised). How they differ, one example of each, and what kind of value each predicts (a category or a number

- Classification — predicting a category

Classification is when a machine learning model learns from past examples to predict which category something belongs to.

- Regression — predicting a number

Regression works in a similar way, but instead of predicting a category, the model predicts a numerical value.

B5. Explain clustering and dimensionality reduction (both unsupervised).

Clustering — grouping similar data

- Clustering is an unsupervised machine learning technique that groups similar data points together based on their characteristics.

B6. For each scenario, say whether it is supervised or unsupervised, then classification, regression or clustering, and why.

(a) Predicting a house's selling price from its size and suburb: This is supervised learning – regression because the model learns from houses with known selling prices and predicts a numerical value, which is the house's price.

(b) Grouping customers by spending habits, with no groups defined upfront: This is unsupervised learning – clustering because the model discovers groups of customers with similar spending habits without being given predefined groups.

(c) Deciding whether an SMS is spam: This is supervised learning – classification because the model learns from SMS messages labelled as spam or not spam and predicts a category for a new message.

(d) Predicting a learner's final mark from term marks: This is supervised learning – regression because the model learns from previous term marks and known final marks to predict a numerical final mark.

(e) Finding groups of products with similar sales patterns: This is unsupervised learning – clustering because the model identifies products with similar sales patterns without being given predefined product groups.

B7.

B3: Comparison of Supervised and Unsupervised Machine Learning

- Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data and use those patterns to make predictions or discover useful information. The two main types of machine learning are supervised learning and unsupervised learning. They differ in the type of data they use and the tasks they perform.

- Supervised learning uses labelled data, which means the data contains examples with known answers. The model learns the relationship between the input features and the correct answers, then uses what it has learned to make predictions about new data. It is commonly used for classification and regression. For example, a school can train a model using learners' study hours, attendance and previous results to predict whether a learner will pass an examination.

- Unsupervised learning uses unlabelled data, which means there are no known target answers provided to the model. Instead, the model searches for patterns, similarities or groups within the data. It is commonly used for clustering and dimensionality reduction. For example, a supermarket can group customers according to their spending habits without deciding the customer groups in advance

- Semi-supervised learning uses a small amount of labelled data together with a larger amount of unlabelled data. It is useful when labelling data is expensive or time-consuming. For example, a hospital may have a small number of medical images labelled by specialists and thousands of

unlabelled images. A model can use both sets of images to help learn patterns for identifying possible diseases.

- Reinforcement learning involves an agent learning through interaction with an environment. The agent takes actions and receives rewards or penalties, learning over time which actions produce better results. For example, a robot can learn to navigate a room by receiving rewards for reaching its destination and penalties for bumping into obstacles. Unlike supervised learning, it does not rely on a dataset of correct answers for every action.

Part C of 4 Preparing the dataa

C1. Explore and Clean

Exploring and cleaning data is the first step in preparing a dataset for machine learning. I use df.info() and df.describe() to understand the dataset, df.isnull().sum() to identify missing values, and df.duplicated() to identify duplicate rows. I use df.drop_duplicates() to remove confirmed duplicate records and astype() or pd.to_numeric() to correct data types when necessary. After cleaning, I check the dataset again to confirm that the changes were successful.

C2. Handle Missing values

Missing values can be handled by removing rows or columns or replacing missing entries with the mean, median, mode or a constant. The appropriate method depends on the type of data, the number of missing values and the presence of outliers. Dropping rows can remove valuable information, especially in small datasets. The median is often preferred over the mean when outliers are present because it is less affected by extreme values. Pandas provides `dropna()` and `fillna()`, while scikit-learn provides `SimpleImputer` for applying reusable imputation strategies.

C3. Encode categorical data

Encoding converts categorical text into numerical values that machine learning models can use. Label encoding is commonly used for target labels, ordinal encoding is appropriate for ordered categories, and one-hot encoding is suitable for unordered categories. Choosing the correct encoder matters because assigning arbitrary numerical values can mislead a model about relationships between categories. One-hot encoding can also create too many columns when a feature contains hundreds of unique categories.

C4. Scale numeric features

Feature scaling adjusts numerical features to comparable scales. Standardisation uses the mean and standard deviation, while Min–Max scaling usually converts values to a range between 0 and 1. RobustScaler uses the median and interquartile range, making it less sensitive to outliers. Scaling is important for algorithms such as KNN, logistic regression and SVM, while decision trees and random forests generally do not require it.

C5. Split into training and test sets

1. Why do we split the data?

When we build a machine learning model, we usually divide our dataset into two parts:

- Training set: The data the model learns from.
- Test set: The data we keep aside to check how well the model predicts information it has not seen before.

What do random state and stratify do?

- random state controls the random selection of rows. Using the same number, such as 42, makes the split reproducible, so you can get the same split again when the data and settings have not changed.
- stratify helps maintain the same approximate proportion of each target category in the training and test sets. It is useful for classification problems, especially when one category is more common than another.

Training and test sets are used to train and evaluate a machine learning model. A common split is 80% for training and 20% for testing. The training set teaches the model to recognise patterns, while the test set checks how well the model predicts unseen data. The random_state parameter makes the random split reproducible, while stratify helps maintain the proportions of target classes in both sets.

Data leakage occurs when information from the test data influences model training or preparation, leading to an unreliable evaluation. To prevent this, the data should be split before scaling. The scaler is fitted on the training data only and then used to transform both the training and test data. In scikit-learn, fit() learns information, transform() applies the learned information, and fit_transform() performs both operations together.

C6.

Splitting Data into Training and Test Sets

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C7. Engineer features

Feature engineering is the process of creating new features from existing data to help a machine learning model make better predictions. Two examples are extracting the month from a date and calculating a patient's age from their year of birth. A correlation matrix helps identify relationships between numerical features. If two features are highly correlated, they may contain similar information, so we can investigate whether both are needed. However, correlation alone should not determine which features to keep or remove because a feature may still be useful for prediction.

Part D of 4

D1. Overfitting and underfitting

Overfitting occurs when a machine learning model learns the training data too closely, including noise, causing it to perform poorly on unseen data. It can be caused by an overly complex model or insufficient training data. Underfitting occurs when a model is too simple to learn the important patterns, resulting in poor performance on both training and test data.

A train/test split helps identify these problems by allowing us to compare performance on data used for training with performance on unseen data. Overfitting is suggested by good training performance and poor test performance, while underfitting is suggested by poor performance on both sets. A model that performs well on both sets, with a small performance gap, is more likely to generalise well.

D2. Measuring a model

Model evaluation is the process of measuring how well a machine learning model makes predictions. For classification, accuracy measures the percentage of correct predictions, precision measures how many predicted positive cases are actually positive, recall measures how many actual positive cases the model identifies, and the F1-score balances precision and recall. A confusion matrix shows true positives, true negatives, false positives and false negatives.

For regression, MAE measures the average absolute prediction error, RMSE penalises large errors more heavily, and R^2 measures how well the model explains variation in the target values.

If only one in every 100 transactions is fraudulent, accuracy is a poor measurement because a model can predict every transaction as legitimate and still achieve 99% accuracy while detecting no fraud. I would prioritise recall if catching fraud is the main goal, while also monitoring precision, F1-score, the confusion matrix and PR-AUC. This gives a clearer picture of whether the model is actually useful for fraud detection.

D3. Cross-validation

K-fold cross-validation is a technique used to evaluate a machine learning model by dividing a dataset into K groups, called folds. The model is trained using $K-1$ folds and evaluated on the remaining fold. This process is repeated until every fold has been used for validation, and the scores are averaged.

Cross-validation is generally more reliable than a single train/test split because the model is evaluated on several different portions of the dataset. This reduces the risk of judging a model based on one unusually easy or difficult split. However, it takes longer to run because the model is trained multiple times.

D4. Parameters vs Hyperparameters

Parameters are values that a machine learning model learns from the training data, while hyperparameters are settings chosen before training to control how the model learns. For example, a decision tree learns its decision rules from the data, while `max_depth=5` is a hyperparameter that limits the depth of the tree.

`GridSearchCV` automatically tests different combinations of hyperparameters and uses cross-validation to identify the combination with the best score according to the selected metric. It can be slow because every combination must be evaluated across multiple folds, which requires training the model many times.

D5. Decision Trees and Random Forests

A decision tree makes predictions by asking questions about the features in a dataset and following branches according to the answers until it reaches a final prediction. It learns its decision rules from the training data.

A random forest usually performs better than a single decision tree because it combines the predictions of many trees. This can reduce overfitting and improve performance on new data, although it is not guaranteed to be better every time.

Neither decision trees nor random forests normally require feature scaling because they split data using thresholds and comparisons rather than distance calculations. Therefore, differences in the numerical scales of features generally do not affect their predictions in the way they can affect distance-based models such as KNN.

D6. K-Means Clustering

K-Means is an unsupervised machine learning algorithm that groups similar data points into clusters. First, the user chooses the number of clusters, K. The algorithm initialises the cluster centres, assigns each data point to its nearest centre, recalculates the centres using the averages of the assigned points, and repeats these steps until the clusters stabilise.

The elbow method helps select K by running the algorithm with different numbers of clusters and plotting the inertia for each value. The best candidate is often the point where the reduction in inertia begins to slow down significantly.

D7.

Class imbalance occurs when one class in a dataset contains significantly more examples than another class. It can mislead machine learning models because they may favour the majority class and achieve high accuracy while failing to identify the minority class.

Class imbalance can be addressed through resampling, class weights and SMOTE. Oversampling increases minority-class examples, while undersampling reduces majority-class examples. Class weights make mistakes on selected classes more costly during training. SMOTE creates synthetic examples of the minority class using existing examples and their neighbours.

D8. Machine learning vs deep learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions. Deep learning is a type of machine learning that uses neural networks with multiple layers to learn complex patterns.

A neural network is a model made of connected units that process information and adjust internal weights during training to improve predictions. For example, a neural network can learn to recognise cats in pictures by learning patterns from many labelled images.

D9. Pipelines

A Pipeline in scikit-learn connects preprocessing steps and a machine learning model into one process. For example, it can fill in missing values, scale numerical data and then train a model. A ColumnTransformer applies different preprocessing steps to different columns. For example, numerical columns can be scaled, while categorical columns can be converted into numerical values using one-hot encoding.

These tools help prevent data leakage and ensure that the same preprocessing steps are used during training and testing. To avoid leakage, the data should first be split into training and testing sets. The pipeline is then fitted only on the training data, allowing it to learn values such as the average or median used to fill missing values. When testing the model, the pipeline reuses those learned values instead of learning from the test data. This helps produce a fairer evaluation and prevents preprocessing steps from being forgotten when making predictions on new data.

D10. Bias and Fairness in Machine Learning

Bias and unfairness can occur in machine learning even when developers do not intend to discriminate. This is because a model learns patterns from its training data. If the data contains historical discrimination or does not fairly represent different groups of people, the model may learn and repeat those patterns.

A documented example is Amazon's experimental AI hiring tool, reported by Reuters in 2018. The tool was trained using approximately ten years of CVs submitted to the company, most of which came from men. As a result, it learned patterns that favoured male applicants and penalised some CVs containing the word “women's.” It also downgraded graduates of two women's colleges. Amazon eventually abandoned the project because it could not guarantee that the tool would make fair hiring decisions

D11. Data Protection and POPIA

POPIA stands for the Protection of Personal Information Act 4 of 2013. It is a South African law that protects personal information and regulates how organisations collect, store, use and share it. When using personal information to train machine learning models, organisations must have a lawful basis for processing the data, explain the purpose of collecting it, collect only necessary information and protect it against unauthorised access. They must also respect the rights of the people whose information they use.

Personal data is information that identifies a person or can reasonably be linked to them, such as a name, identity number, telephone number or identifiable financial records. Anonymised data is information that has been changed so that individuals can no longer reasonably be identified. For example, a bank's customer records containing names and account details are personal information, while a sufficiently broad summary showing the number of customers in different income groups may be anonymised. However, removing names alone does not guarantee anonymity.

Presentation and referencing:

I used ChatGPT to help me understand machine learning concepts, explain technical terms in simple language, develop examples, structure my assignment and improve the clarity of my writing. I used the tool as a learning aid while working through the assignment questions. I am responsible for reviewing the final work, checking the accuracy of the information and ensuring that the submitted assignment reflects my understanding. I have identified external sources in the reference list where applicable.